Combining expressions

Substitute an expression and collect like terms.

Name: _____ Suggested time: 10 minutes

Show your working. Substitute carefully and simplify.

01 Given a=2b+2

Write in terms of b and simplify

Simplify: 5a+4b

02 Given a=3b+5

Write in terms of b and simplify

Simplify: 2a+5b

03 Given a=6b+1

Write in terms of b and simplify

Simplify: 2a+6b

04 Given a=2b-5

Write in terms of b and simplify

Simplify: 3a+6b

Answer sheet

Combining expressions | 4 questions

Keep this section separate from the question sheet. Full working follows on the next pages.

01 $14b+10$

02 $11b+10$

03 $18b+2$

04 $12b-15$

Worked solutions

Combining expressions

01 Given a=2b+2

5a+4b

Substitute a=2b+2.

5(2b+2) + 4b

10b +10 + 4b

(10+4)b +10 = 14b+10

02 Given a=3b+5

2a+5b

Substitute a=3b+5.

2(3b+5) + 5b

6b +10 + 5b

(6+5)b +10 = 11b+10

03 Given a=6b+1

2a+6b

Substitute a=6b+1.

2(6b+1) + 6b

12b +2 + 6b

(12+6)b +2 = 18b+2

04 Given a=2b-5

3a+6b

Substitute a=2b-5.

3(2b-5) + 6b

6b -15 + 6b

(6+6)b -15 = 12b-15